

Loopscale Exponent LP Pricing Audit

Executive Summary

Introduction

Offside Labs completed a security audit of Loopscale smart contracts, starting on Oct 30, 2025, and concluding on Oct 31, 2025.

Project Overview

The audit concerns the Loopscale protocol on the Solana blockchain, a modular, order-book-based lending platform that enables over-collateralised borrowing and lending across a wide variety of digital-asset collateral — including tokenised liquidity-provider (LP) positions, staked assets and other advanced DeFi primitives.

The audit was conducted **at request of Exponent Finance** and focuses solely on the pricing mechanism for Exponent LP tokens when used as collateral within the Loopscale lending markets. Our review is limited to the portion of code responsible for calculating and assigning the collateral value of Exponent LP tokens.

The scope of this review solely covers the **pricing logic related to Exponent LP tokens**. It does **not** represent a comprehensive audit of the Loopscale protocol. The report excludes review of the broader Loopscale codebase, other collateral types, loan matching mechanisms, and liquidation processes.

Audit Scope

The assessment scope contains mainly the smart contracts of the Exponent-LP-Pricing program for the Loopscale project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- Loopscale
 - Codebase: <https://github.com/LoopscaleLabs/loopscale-program-library>
 - Commit Hash: [b60bc63ea9c52d3f425e41f356b3100434350f6d](#)

We listed the files we have audited below:

- Loopscale
 - `programs/loopscale/src/state/oracle.rs`
 - `programs/loopscale/src/utls/oracle/discount_to_maturity.rs`
 - `programs/loopscale/src/utls/oracle/exponent_lp.rs`
 - `programs/loopscale/src/utls/oracle/exponent.rs`

Findings

The security audit revealed:

- 0 critical issues

- 0 high issues
- 1 medium issues
- 0 low issues
- 0 informational issues

Further details, including the nature of these issues and recommendations for their remediation, are detailed in the subsequent sections of this report.

Summary of Findings

Key Findings and Recommendations

▼ Incorrect LP Value Calculation Due to Missing SY Exchange Rate

- Severity: Medium
- Target: Smart Contract
- Category: Logic Error
- Status: Fixed

Description

In the `get_exponent_lp_token_price_with_quantity` function, the value of the SY token component of the LP token is calculated as `underlying_token_value_per_lp_token`. This value is derived by dividing `sy_balance` by `lp_supply`. This calculation incorrectly assumes a 1:1 exchange rate between the SY token and the underlying token of the market. However, within the Exponent protocol, the relationship between an SY token and its underlying token is defined by `sy_exchange_rate`, where 1 SY token can be redeemed for `sy_exchange_rate` units of the underlying token. The current implementation does not apply this exchange rate, leading to a miscalculation of the LP token's value when the `quote_mint` (the market's underlying token) is not the SY token itself.

```
let underlying_balance = Decimal::new(convert_u64_to_i64(exponent_market_data.sy_balance)?, underlying_mint.decimals.into());
let lp_supply = Decimal::new(convert_u64_to_i64(lp_mint.supply)?, lp_mint.decimals.into());
let quantity_dec = Decimal::new(convert_u64_to_i64(quantity)?, lp_mint.decimals.into());

let underlying_token_value_per_lp_token = underlying_balance.checked_div(lp_supply).ok_or_else(arithmetic_error!());
```

Impact

The failure to incorporate the `sy_exchange_rate` results in an inaccurate valuation of the SY portion of the LP token. This can lead to the oracle reporting a flawed price for the LP token.

Recommendation

It is recommended to modify the function to fetch the current `sy_exchange_rate` for the market. The `underlying_token_value_per_lp_token` calculation should be adjusted to:

```
let underlying_token_value_per_lp_token = (sy_balance * sy_exchange_rate) / lp_supply;
```

This will ensure that the value of the SY component is accurately reflected in terms of the underlying token, providing a correct and reliable price for the LP token.

Mitigation Review Log

Fixed in commit 77867c211572288edc896a51e57dc7586e954519.